Methodology report: Calibrated Everywhere

Paper: “Calibrated Everywhere: A Dilemma for Track-Record Reliabilism about Credence” (AI Philosophy Competition, Paper 1, Stage 2).

This report reconstructs the method from the running log (`log.md`) and the numbered files in this folder. The full chat logs of the session are retained and can be inspected; every subagent prompt quoted below is reproduced from them.

1. Division of labour

Human inputs. These stayed within what the competition permits:

1. Choice of the area (formal epistemology: reliabilism about justified credence).
2. Selection of the thesis from candidates generated by an AI in Stage 1 (“calibrationist reliabilism fails because, by the Foster-Vohra theorem, a process with no information about the world can be calibrated in every possible world; generalization via the expert-testing literature”).
3. A generic step-by-step method instruction for Stage 2, with generic ground rules: numbered files, no overwriting, explicit premises, no unverified attributions, a running log. The steps were a novelty check, mathematics, a steelman, three rounds of outlines, a draft, an independent referee panel, a citation audit, final checks and this report.

The only other message was operational: a request to resume two referee subagents after an API rate limit. No human wrote, edited, selected or commented on any argument, outline, draft, referee report or revision. The full chat logs are retained.

AI work (everything else): literature search and reading, statement of the mathematics, the steelman, the outlines and critiques, all drafts, the anti-tracking lemma and the other own results, the design and prompting of the referee subagents, the revisions in response to them, the audit, the PDF production and this report. One AI agent (the “author agent”) did the main work. It launched fresh subagents for the referee panels and the audit.

2. Environment constraints that shaped the method

- **No full-text access.** The environment’s network policy blocked every scholarly host (arXiv, PhilPapers, PhilArchive, SEP, Springer, Cambridge, JSTOR, faculty pages). Only a web-search tool returning summaries and verbatim snippets was available, to the author agent and to every subagent.

- **Consequence 1.** Every claim about what an author holds was restricted to what abstracts, snippets or reliable secondary descriptions showed. These are recorded in `sources/search_notes_step1.md`, `sources/search_notes_step2.md` and `sources/search_notes_step6.md`.
- **Consequence 2.** Quotations whose exact wording could not be confirmed were paraphrased. For example, a sentence from Olszewski and Sandroni 2008 that one round-4 referee found worded differently in another version of the abstract was turned into a paraphrase.
- **Consequence 3.** Primary-text exegesis of Dunn (2015), Tang (2016) and Goldman (1986) is limited. Several referees asked for page-referenced engagement with those texts, and this could not be supplied.
- **Repository.** For most of the session, pushing to the remote repository was refused, so work was committed locally. It was pushed once access had been restored.

3. The steps

Step 1. Novelty check (`01_novelty.md`)

- About 25 searches: the target papers (Dunn 2015, Tang 2016, Pettigrew 2021, the SEP entry, Dawid 1982, Seidenfeld 1985, Belot 2013), philosophy papers citing Foster and Vohra (1998) or Sandroni (2003), and calibrationism variants.
- **Result.** No paper found that makes the argument.
- **Precursors identified:** Goldman's 70% case and the nearby-worlds reply (as reported by the SEP), Tang's evidence-insensitivity objection, Seidenfeld's "calibration in the long run is otiose", and Belot on the typicality of calibration failure.
- **Correction to the Stage 1 thesis.** The generalization "no outcome-based test separates informed from uninformed" is false as stated, because non-manipulable tests exist. It was restricted in Step 2.

Step 2. Mathematics (`02_math.md`)

- **Statements.** Each theorem was stated with its assumptions, checked against abstracts and snippets, and marked with a verification level (abstract-level, secondary, or own result). The results covered:
  - Foster-Vohra;
  - deterministic smooth calibration (Foster-Hart);
  - checking rules and multicalibration;
  - manipulability (Sandroni; Olszewski-Sandroni; Shmaya);
  - the limits of manipulability (Dekel-Feinberg; Olszewski-Sandroni 2009b; Fortnow-Vohra; Al-Najjar et al.; Al-Najjar-Weinstein).
- **Own elementary results.** The anti-tracking lemma; its combination with calibration; the generalization; dominance.

- **List of claims the paper must not make**, for example “calibrated in every possible world” without qualification.

Step 3. Steelman (`03_steelman.md`)

- **Variants.** Seventeen versions of calibrationism, including several not stated in the literature (multicalibration, comparative, non-manipulable-test reliabilism, computationally bounded, chance given grounds). Each was checked against Step 2.
- **Survivors.** The chance-given-grounds view (to be accommodated as the positive view), non-manipulable-test reliabilism (to be stated as a limit), and comparative calibrationism.

Step 4. Outlines (`04a`, `04b`, `04c`)

Three rounds, each followed by a written critique and a revision. The main structural change came in round 2: the core argument was recast as a schematic argument about standards that a single process can meet in every world, with the barometer case demoted to an illustration.

Step 5. Draft (`05_draft_v1.md`)

First full draft, about 5,270 words, with the core argument as nine numbered steps.

Step 6. Referee panel (`referee/`, `06a`, `06b`, `06c`, drafts `05_draft_v2` to `v5`)

Design. - Each round had three fresh subagents. - Each saw only a copy of the current draft, placed in a scratchpad directory outside the project folder, and was told not to open any other file. - Roles: - A: a formal epistemologist working on accuracy and scoring rules. - B: a reliabilist sympathetic to Dunn, instructed to defend calibrationism as hard as possible. - C: an economist or theoretical computer scientist checking every technical claim. - Each referee produced a verdict and scores (originality, correctness, importance). - From round 3, referees labelled each point substantive or presentational.

Stopping rule given. Two consecutive rounds raising no new substantive objection, or four rounds in total. The first condition was never met. The panel stopped at the four-round cap.

Rounds.

Round	Draft	A (O/C/I)	B (O/C/I)	C (O/C/I)	Verdicts
1	v1	7/5/7	8/5/7	7/5/7	all major
2	v2	7/5/8	7/5/7	7/6/7	all major

Round	Draft	A (O/C/I)	B (O/C/I)	C (O/C/I)	Verdicts
3	v3	7/5/8	8/6/8	6/7/6	major, major, minor
4	v4	6/6/7	7/7/7	6/7/6	major, major, minor

What changed after each round.

- **After round 1 (v2).** The thesis became a dilemma.
  - Horn 1: standards that one uninformed process can meet in every world (“securable” standards) are insufficient for reliability.
  - Horn 2: standards that cannot be secured are insufficient, or owe their force to agreement with the chances.
  - The lemma now integrated over coin flips properly. A run-level version was added (probability at least 2/7).
  - Theorem hypotheses were corrected: finite-time rejection in Olszewski-Sandroni 2008; probability-one passing in Shmaya 2008.
  - Mischaracterizations of Olszewski-Sandroni 2009b and Al-Najjar-Weinstein were corrected.
  - Rates were corrected and the target was restated with reflexivity at the actual world.
- **After round 2 (v3).**
  - A false appendix claim (A3) was corrected. All three referees had independently found the same counterexample: a process that flips one coin at the outset.
  - Calibration in a world was regimented as an almost-sure property.
  - Premise (R) gained a resources clause, to answer the charge that it was vacuous.
  - Normal-worlds calibrationism was declared a limit of the argument.
  - The demon-world discussion was corrected.
  - The Hu-Shmaya computability exception was added.
- **After round 3 (v4).**
  - An abstract was added.
  - The deterministic witness was extended to finitely many reference classes (own construction).
  - “Same resources” was replaced by computability relative to the process’s rule.
  - A faulty argument (the old A4) was dropped.
  - Computable calibrationism was stated as not refuted by the first horn. An initial-segment objection was added, and the conclusion against track-record reliability was made conditional on a non-reductive view of chance.
  - The pass-probability thresholds in (G) were made explicit.
- **After round 4 (v5, final).**
  - The exact scope of the first horn was stated, and unrestricted computable calibration was reclassified onto it.

    - Horn 1 was split into a deterministic core (feasible, ordinary world) and a randomized extension (contestable reading).
    - The statement of Shmaya's theorem was restricted to sequential forecasts.
    - The rate claims were corrected.
    - A sub-dilemma for computable calibrationism was added.
    - Outcome indistinguishability was discussed.
    - The positioning relative to Pettigrew and Clifton was revised.
    - The thesis was restated as: track-record calibrationism fails, and calibrationism read in terms of chances is the chance view.

Points the paper concedes or leaves open in its final form, following the referees: - the normal-worlds version of calibrationism (aggregate reliability across normal worlds); - the dependence of the anti-(D2) conclusion on the metaphysics of chance; - the reliance of the randomized extension on a computability rather than a feasibility reading; - the reference-class problem for coarse-grained chance in the positive view; - the lack of primary-text exegesis.

Step 7. Citation and attribution audit (`07_audit.md`)

A fresh subagent that saw only the final draft checked every bibliographic entry, every attribution and every quotation against sources reachable by web search. It classified each as verified, partly verified, not verified or wrong. The fixes are listed in `07_audit.md` and applied in `05_draft_v6.md`. The auditor verified 36 of 40 references and most in-text claims at abstract or snippet level; it flagged unverifiable quotations, a misattribution (Goldman 1979 vs 1986), a missing citation (Dawid 1985b), a misplaced source for the Brier decomposition, the continuum-hypothesis caveat for Dekel-Feinberg, separate attribution of the lower bounds, and several reference details. All fixes were applied except two items that could not be verified either way (page numbers for Qiao-Valiant and Goldman 1986), which were left without pages, and the Oakes report of Dawid, which was kept on the strength of the step-6 search notes (see `07_audit.md`).

Step 8. Final checks (`08_final_checks.md`, `final.pdf`)

Word count 5,744 (plain text, excluding bibliography; other counting methods give at most 5,977). Anonymization, hidden-text and rendering checks passed; details in `08_final_checks.md`. Output: `final.pdf`.

4. Reproducibility

- Every step's output is a numbered file. No file was overwritten; new versions received new names (`05_draft_v1` to `v6`; this report itself was versioned as `09_methodology_report_v2.md`).
- The referee reports are condensed records written by the author agent. All substantive points are retained, and the full reports are in the retained chat

logs.

- Search notes record the evidence for every attribution.
- The referee prompts are role prompts of the form given in Step 6. Each instructs the referee to read only the submission file, to write a report with a verdict and scores, and, from round 3, to label points substantive or presentational.
- To reproduce: run the same pipeline with the same generic instructions, and give the same role prompts to fresh agents that see only the draft.
- One environmental feature should be noted. A post-processing step in the environment re-wrapped the wording of files after they were written; the files were reported as “changed on disk”. The author agent re-read the files after each such change and worked from their current state.

5. Process interruptions

The session was interrupted twice by usage limits (once during the first referee round, once during the final checks). After each, the user wrote only that the limit had been reset and asked the agent to continue; these messages contained no content about the paper.